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Computer Engineering Y3 2026

Mobile Application Development (Flutter)

Posts

This is aflutter based app that manages post. This app integrates the JSON Placeholder test API to allow crud operation on posts.

This app allow staff users to view all post, open post details edit existing post and even delete them

Building this app the following dependencies were used

1. flutter sdk: allowed me to build complete UI, navigation forms, state handling and Material components
2. http package : this allowed communication of my frontend with the REST APIs for CRUD operations.
3. flutter lints : these helped me to keep code style and quality.

Api Exception Handling

I handled exception both in UI and service layer

* service : - checked status code for every request
  - Threw exceptions for unexpected status codes
  - validated JSON format before parsing.

* UI : - try and catch on async actions
  - Displayed snack bar for feed back and Error widgets
  - Used loading indicators (req in progress)
  - Used mount check before updating state after async calls

This allowed to handle common errors like network failure invalid response and other error while keeping the app stable

CamScanner

Future and Future Builder

I used future for all synchronous API operations and Future Builder to build the posts list UI based on the result of fetch operation.

UI parts that depends on future Builder are

* loading state : show progress while waiting for response
* Error state : Show error message & retry button if request fail.
* Success state : Show list of posts when data is loaded
* Empty state : It would show message when no post available

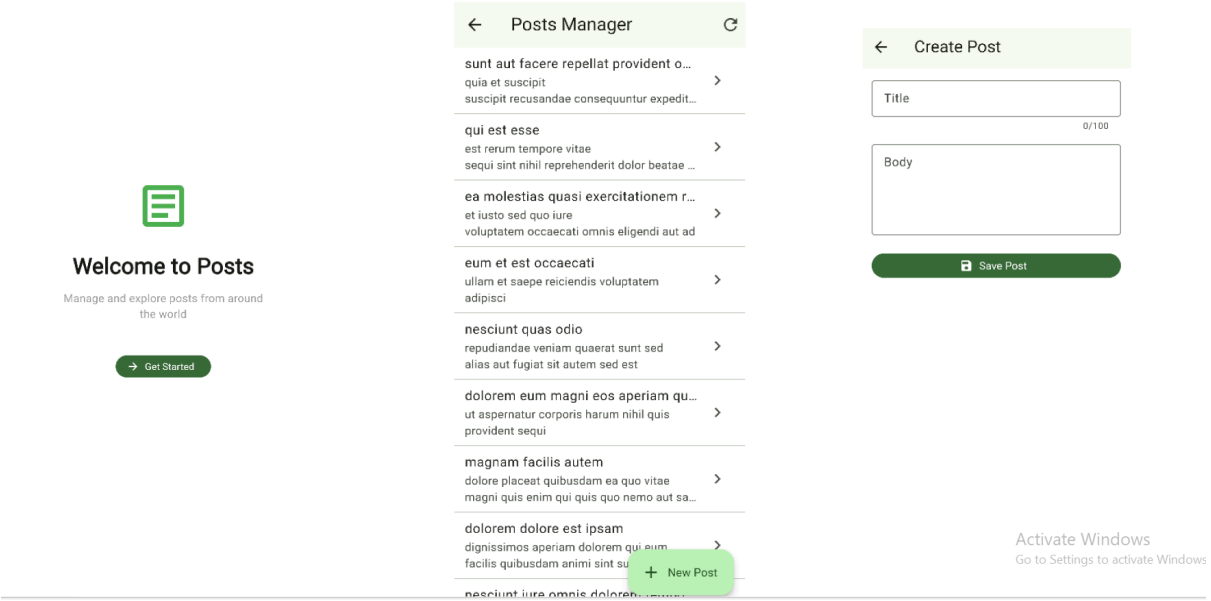
Future and future Builder made UI reactive thus believing it improve user experience

Generally This project allowed me to learn how to integrate backend APIs in mobile app, interact with async handling with Future and Future Builder and do proper Error handling.

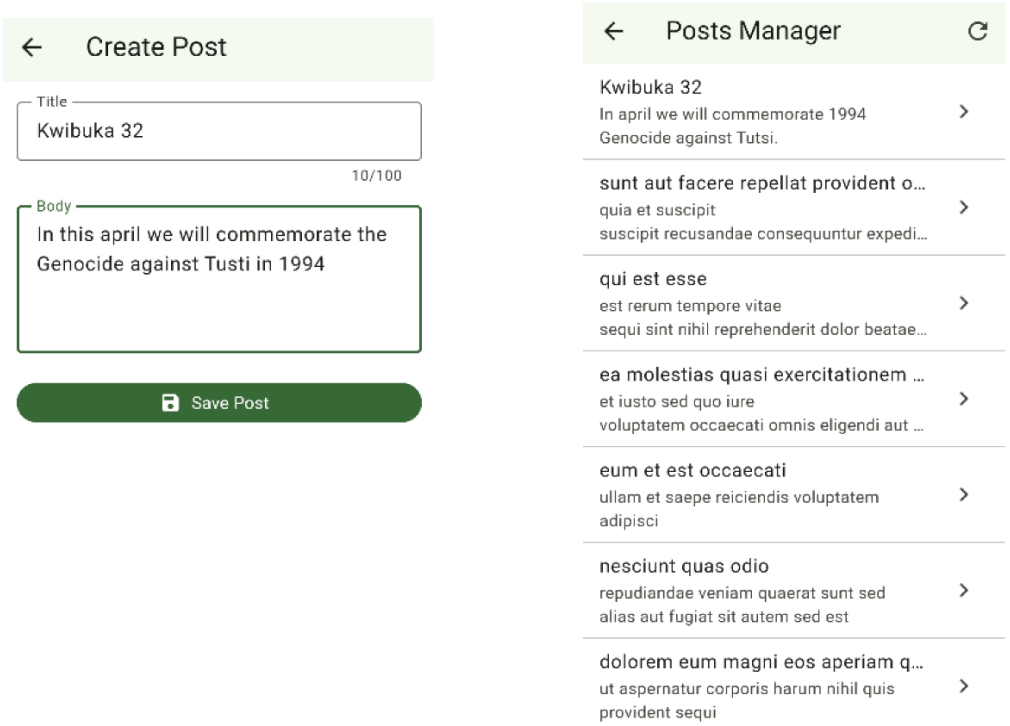
CamScanner

Screen shots of the posts app: I aligned 3 screens in one screenshot

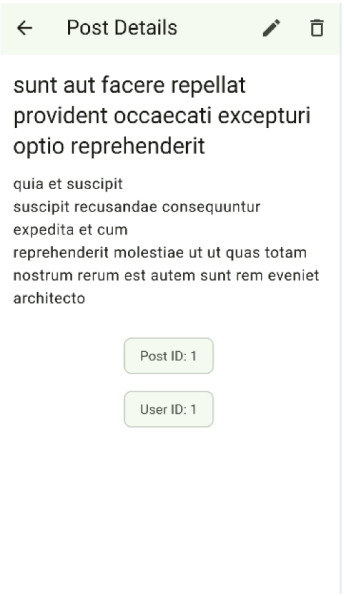
When you do flutter run you land on Welcome page -> Get started button ->Floating button to create new post

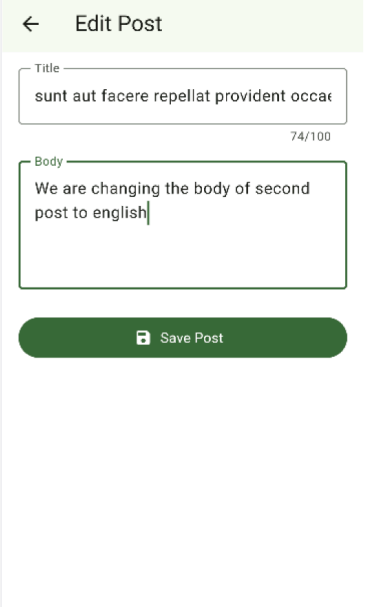


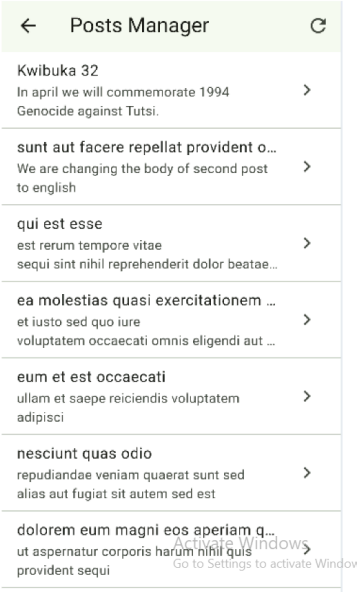
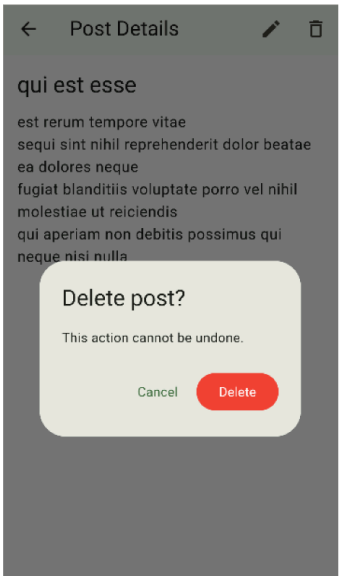
Adding info of the post



I choose to edit the second post: whe n you choose one of the post the post details screen come with different info







Delete Process: I choose to delete the third post